Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	*(a)*		Moment of force about a point = Force × <u>perpendicular distance from [**not** around] the point</u> (to the line of action of the force). Accept symbol for perpendicular	1			1		
	(b)	(i)	Clockwise moment [CM] = 52 × 0.15 **Or** CM = 7.8 (N m) seen		1		1	1	
		(ii)	F × 0.58 = 8 **or** answer to (b)(i) **or** F × 0.58 =52 × 0.15 (**ecf** from (b)(i)) (1) F = 13.8 N (or 13.4 if 7.8 N m used – accept 13.5) (1) [Accept 2 s.f.] Correct answer → 2 marks		2		2	2	
	(c)		In position 2 [perpendicular] distance of weight from hinge is smaller (1) …. so CM decreased (1) So ACM reduced (1) so force in bar decreased and so Tom incorrect **o**r Bethan correct (1) **Or** (converse argument): In position 1 [perpendicular] distance of weight from hinge is greater (1)… so CM increased (1) So ACM increased (1) So force in bar increased and so Tom incorrect or Bethan correct (1) Accept answers based on calculation Do not accept reference to Tom/Bethan being correct/incorrect without explanation Alternative explanation using vertical position of bar: In position 1 the bar is closer to the pivot (1) .. so to balance the clockwise moment (1).. the force in bar is increased so Tom incorrect / Bethan correct (1)			4	4		
1			**Question 1 total**	**1**	**3**	**4**	**8**	**3**	**0**